

Seoyun Baek

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Research Interest | *Machine Learning, Computer Vision*

Education

- Korea Advanced Institute of Science & Technology (KAIST)**, Daejeon, South Korea *Feb 2025 – Present*
- B.S. Industrial & System Engineering
- Korean Minjok Leadership Academy**, Gangwon-do, South Korea *Mar 2022 – Feb 2025*
- High School Diploma

Publications and Conferences

- Seoyun Baek** (2024). "Dynamic Asset Allocation with Markowitz Theory and Proximal Policy Optimization," *KNOWLEDGE TO LEAD* (ISSN: 2765-0626), Vol. 5, No. 1, pp. 17-22.
- Seoyun Baek**, Jungwoo Brian Kim, Jihoon Oh (2024). "Emotion Recognition Device for Visually Impaired," *International Journal of STEAM*, Vol. 15, No. 1, pp. 80-83.

Work Experiences

Vision and Learning Laboratory @ KAIST CS, Daejeon, South Korea *Jun 2026 – Present*
Research Intern

- Researching Computer Vision under the supervision of Prof. Seunghoon Hong.

PeoplesLeague, Daejeon, South Korea *Mar 2025 – Jul 2025*
Research Intern, Software Department

- Organized and led a data labeling team to build a high-quality annotated dataset for training the cooking-robot segmentation model.
- Developed a segmentation model for cooking robots to accurately identify safety-critical objects (pots, ingredients, oil, and human operators) in real-time.
- Designed a custom evaluation metric to prevent real-world hazards by penalizing consecutive misclassifications of high-risk classes.
- Optimized the model based on the custom metric, achieving $\geq 95\%$ safety-compliant frames and limiting maximum consecutive errors to under 10 frames (< 2 seconds).

Selected Honors and Awards

- 3rd prize, **International Youth Research Paper Competition**, HOBY *Sep 2024*
- Among 300+ Papers Submitted Worldwide
- 4th Industrial Revolution POWER KOREA Scholarship for Talented**, National Assembly *Aug 2024*
- Finalist, **International Science & Engineering Fair**, Regeneron & Society for Science *May 2024*
- Selected as representative of South Korea
- Gold Medal, **Korea Science & Engineering Fair**, KSS *Feb 2024*

- 1st place among 300+ Papers Submitted Nationwide

Grand Prize, **KISTI & UNIST Supercomputing Challenge**

Jul 2023

- 1st place among 298 teams nationwide, participated as a team leader

Selected Projects

5th KAIST-POSTECH-UNIST Data Science Competition

Nov 2025 – Dec 2025

(Sponsored by Hankook Tire)

- Developed a machine learning pipeline using Finite Element Method (FEM) simulation and design specification data to predict tire manufacturing defects and optimize test production strategies.
- Engineered custom features, including Target-Aware Correlation Drop and Mean-Difference Flags (MDFlag), and extracted structural statistics (e.g., pressure gradients) from complex 2D FEM data to isolate defect signals.
- Optimized predictive stability on a highly imbalanced dataset (Defect ratio ~15%) by implementing a weighted ensemble model combining TabPFN (neural network for tabular data) and Random Forest.
- Designed a profit-maximizing decision algorithm based on break-even analysis (Good: +100, Defect: -2000) to strictly penalize false negatives, strategically selecting the top 200 optimal tires to maximize Total Net Profit.

Dynamic Asset Allocation with Markowitz Theory and Proximal Policy Optimization

Jul 2024 – Sep 2023

- Addressed the high initial volatility and risk of standalone reinforcement learning models by implementing an ensemble technique that computes a weighted average of MVO and PPO asset distribution weights.
- Conducted backtesting using 10 years of S&P 500 market data, achieving a cumulative return that outperformed the standalone Markowitz portfolio by 168% and the PPO-only portfolio by 336%.
- Demonstrated superior risk-adjusted performance and robust risk management by maximizing both the Sharpe and Sortino ratios compared to traditional baseline models.

Emotion Recognition Device for Visually Impaired

Aug 2023 – Sep 2024

- Engineered a hands-free, glasses-shaped wearable device designed to assist visually impaired individuals by conveying conversational emotions through a custom-coded vibration sensor.
- Developed deep learning-based multimodal emotion recognition algorithms, achieving 98.55% accuracy in Facial Expression Recognition (FER) using the Xception model and 97.25% in Speech Emotion Recognition (SER) via Conv1D and LSTM.
- Implemented an ESP32 microcontroller for real-time hardware activation via Bluetooth Low Energy (BLE) communication, controlled by a custom-built Flutter mobile application.
- Designed an emotion integration logic that categorizes extracted emotions into positive, negative, and neutral states, cross-validating facial and audio inputs to ensure accurate and refined vibration feedback.

Extracurricular Activities and Leadership

Department of Industrial & System Engineering, Representative, KAIST

Mar 2026 – Present

- Directing internal and external departmental policies and oversaw overall student operations and initiatives

VOK (KAIST Broadcasting System), Executive Director of General Affairs, KAIST

Nov 2025 – Present

- Managing entire financial operations, budget planning, and accounting processes for the organization.
- Assisting the President with executive decision-making and overall station administration.

VOK (KAIST Broadcasting System), Announcer, KAIST

Mar 2025 – Present

- Anchored campus news and radio programs, and moderated major university festivals and events.
- Completed professional announcing training.

Entrance Ceremony, Freshman Representative Speaker, KAIST

Feb 2025 - Feb 2025

- Delivered the freshman representative speech on behalf of 800 incoming undergraduate students at the 2025 KAIST Entrance Ceremony.
- <https://drive.google.com/file/d/1h14jmnJU6mmQ-nee50IOw6up1jisG33O/view>

Skills Summary

Programming Languages: Python, C, C++, Java

Frameworks/Tools: PyTorch, TensorFlow, OpenCV, CUDA, NumPy, Pandas, Seaborn

Hardware: Jetson Nano, Arduino, ESP32

Languages: Korean (Native), English (Fluent)